

MA 403 Real Analysis

Homework 7

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, and suppose that $|f(x) - f(y)| \leq (x - y)^2$ for all $x, y \in \mathbb{R}$. Prove that f is constant.

2. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ as follows:

$$f(x) = \begin{cases} e^{-(1/x^2)} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

Prove that:

- (a) f is continuous for all x .
(b) $f^{(n)}$ exists and is continuous for all x , and $f^{(n)}(0) = 0$, $n = 1, 2, \dots$.

3. We say a function $g : (a, b) \rightarrow \mathbb{R}$ is of class C^k if the k^{th} derivative $g^{(k)}$ exists and is continuous on (a, b) .

$$\text{Let } f_n(x) = \begin{cases} x^n \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

Prove that f_1 is continuous but not differentiable at 0. Prove that f_2 is differentiable but not of class C^1 . In general what can you say about f_n ?

4. Let $f : (a, b) \rightarrow \mathbb{R}$ be differentiable and suppose $f'(x) = 0$ for all $x \in (a, b)$. Prove that f is a constant function.

5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that $f'(x)$ exists for all $x \neq 0$ and such that $\lim_{x \rightarrow 0} f'(x) = 3$. Prove that $f'(0)$ exists and $f'(0) = 3$.

Hint: Mean value theorem.

6. Assume f has a finite derivative in (a, b) and is continuous on $[a, b]$, with $a \leq f(x) \leq b$ for all $x \in [a, b]$ and $|f'(x)| \leq \alpha < 1$ for all $x \in (a, b)$. Prove that f has a unique fixed point in $[a, b]$.